

Josh Beira

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EDUCATION

University of Warwick

Bachelor of Science (Honours) in Computer Science

Coventry, UK

Sep 2025 – Jul 2028

Cadbury Sixth Form

*A-levels: A*A*A*A*A* (Mathematics, Further Mathematics, Computer Science, Physics, EPQ)*

Birmingham, UK

Sep 2023 – Jun 2025

EXPERIENCE

Research Intern (Artificial Intelligence and Machine Learning)

Apr 2026 – June 2026

Department of Computer Science, University of Warwick

Coventry, UK

- Awarded a highly competitive university bursary to conduct independent research, demonstrating strong **analytical** skills by designing a more efficient and innovative approach to **Text-to-Image Person Re-Identification**
- Engineered a Generative Data Augmentation pipeline, utilising **Stable Diffusion** to replace image backgrounds and isolate subjects to train and fine-tune **OpenAI's CLIP** model on a highly robust, noise-reduced dataset
- Applied strong mathematical fundamentals to formulate custom **loss functions** that optimise model architecture, and rigorously evaluated performance against academic benchmarks

Head of Machine Learning

October 2025 – Present

Warwick Racing

Coventry, UK

- Utilised **PyTorch**, **TensorFlow**, **OpenCV**, and **ROS** to build a real-time perception system using image segmentation **CNNs** to detect cones. Improved accuracy by **47%** by adding LiDAR clustering
- Led development of **RL** and **supervised learning** steering models, leveraging **SQL** to work with large, complex datasets and reduce steering error by **34%** through analytical thinking and attention to detail
- Built ready-to-use systems using **Docker** containerisation, demonstrated innovation, problem-solving and leadership by working with electrical and mechanical teams to convert vision into reliable ≤ 50 ms steering/speed commands
- Collaborated and communicated effectively in the team through **Agile** collaboration and expert use of **Git** and **GitHub**, and presented technical work clearly through documentation and demos to non-technical team members

Python Tutor

Jan 2026 – Present

software Academy

London, UK

- Applied a **logical**, **analytical**, and **creative** approach to diagnose learning gaps and design customised curricula, maintaining a meticulous approach to managing independent scheduling and my own workload
- Demonstrated **strong communication** by **explaining complex concepts to non-technical audiences** and translating advanced Python and Machine Learning topics into accessible lessons for children and parents

PROJECTS

The Warwick Hackathon (WHACK) | *Flask, SQL, Node, React, JavaScript, HTML, CSS*

October 2025

- Led a 4-person team to build a sophisticated full-stack AI Powered Finance Assistant web application, Financial Literacy and Contract Analysis app, and presented a demo to non-technical judges
- Engineered asynchronous workflows via the **Google API** to orchestrate parallel Gemini 2.5 Pro agents, utilising advanced **prompt engineering** to significantly reduce system latency and collaborating through **Git/GitHub**

COVID-19 Mortality Risk Model | *Java*

June 2025 – September 2025

- Built a logistic-regression mortality risk model **from scratch** without using high level ML libraries, in **Python** using **NumPy**, **Pandas**, **gradient-descent**, **calculus** to estimate patient survival probabilities
- Analysed and cleaned a large clinical dataset found on **Kaggle**, applied **feature-engineering** to select useful features assessed model performance using **confusion matrix**, precision and recall achieving accuracy of **75.82%**

2025 F1 Winner Predictor | *Python*

March 2025 – Present

- Built a **Gradient Boosting regressor** model using **Scikit-Learn** using historical race data from **FastF1 API** to predict race times for 2025 and rank drivers accordingly
- Applied **feature engineering** techniques to improve predictions, performance is measured using **MAE**

TECHNICAL SKILLS

Languages: Python, Java, C/C++, Haskell, Rust, SQL, JavaScript, Assembly, HTML/CSS, Bash

Libraries/frameworks: PyTorch, TensorFlow, Scikit-learn, OpenCV, Pandas, NumPy, Matplotlib, React, Flask

Developer Tools: Git, GitHub, Docker, UNIX utilities, Weights and Biases, AWS, Azure, Google Cloud, n8n